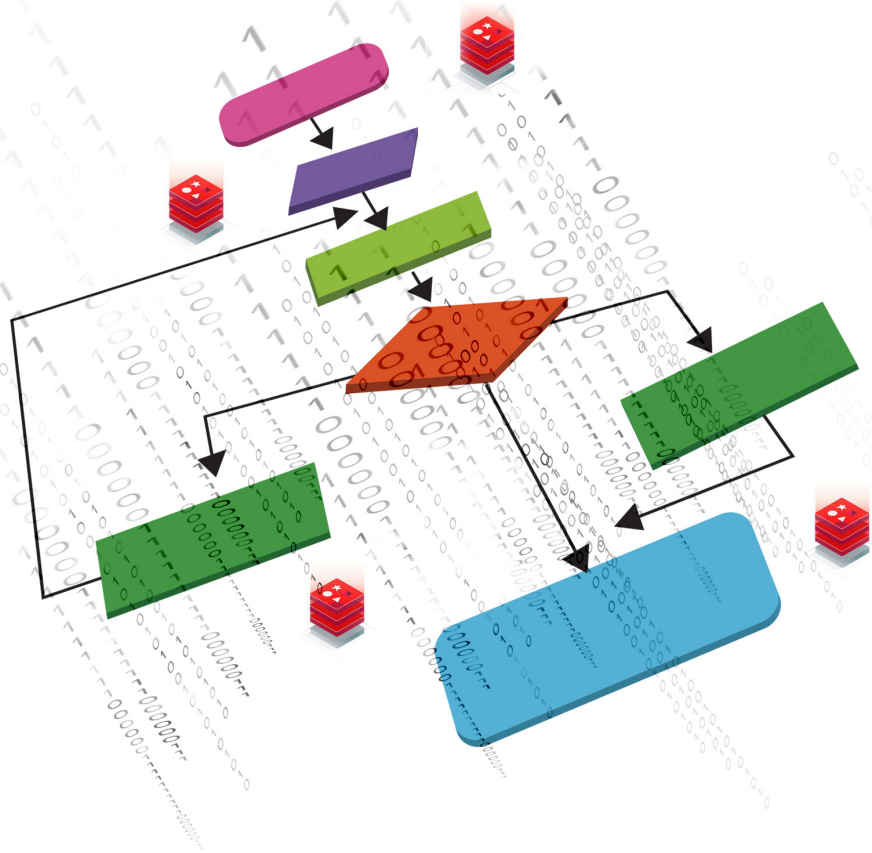


An Introduction to Redis, the In-Memory Data Structure Store

Specialised representation of data helps applications perform better. This article introduces an open source data structure store called Redis, which enables efficient representation of data with various data structures such as Lists, Hashes and Sets. Features like transaction support and persistence on disks make Redis an exciting choice for efficient data handling.



A critical factor that decides the optimal performance of a software application is the ability to handle data using specialised representation. However, optimised the business logic may be, a delay in fetching the data will scale down an application's performance. To make the application perform well, the associated data feeding and representation should be done with the highest possible precision and efficiency. A delay of even a few nanoseconds can cascade into a significant one and make the end user feel the lag in the application.

Redis is one of the many solutions that enable applications to fetch and handle data in an efficient manner with the help of various data structures.

Redis is polymorphic in nature, and may be used in many ways. Some of the important forms in which it can be used are listed below (see Figure 1):

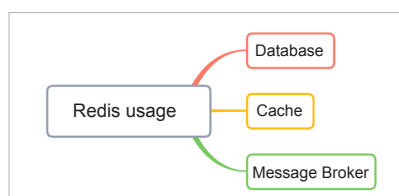


Figure 1: Redis usage

- Database
- Cache
- Message broker

Redis data structures and features

The power of Redis lies in the availability of numerous types of data structures. Each of these enables the representation of data in a specific way, which helps to increase the access speed. The Redis data structures are:

- Strings
- Hashes
- Lists
- Sets